

Homework 6

Please complete the following problems. You must submit the following for each problem: a) the answer you are asked to return and b) Python code that can be run to verify your work.

1. Global Multiple Alignment

Recall problems from children's books where you find the odd object. When you're dealing with nucleotide strings you can face the same problem, like detection of a foreign sequence among a batch of samples. Such analysis nowadays is performed by many countries to find smuggling or fake food (e.g., passing tilapia off as salmon; see Figure 1).



Figure 1: Fish samples waiting for analysis in the lab of US Customs and Border Protection (Getty Images).

To compare several homologous strings, we need to align all of them simultaneously, a procedure known as multiple sequence alignment, or MSA. Because it requires you to compare more than two sequences at once, MSA is a more complicated problem than pairwise alignment. In fact, finding an optimal alignment between more than a very few sequences is so computationally intensive that many MSA programs rely instead on "quick and dirty" heuristic methods that are guaranteed to provide a "good" solution but not necessarily the best possible one.

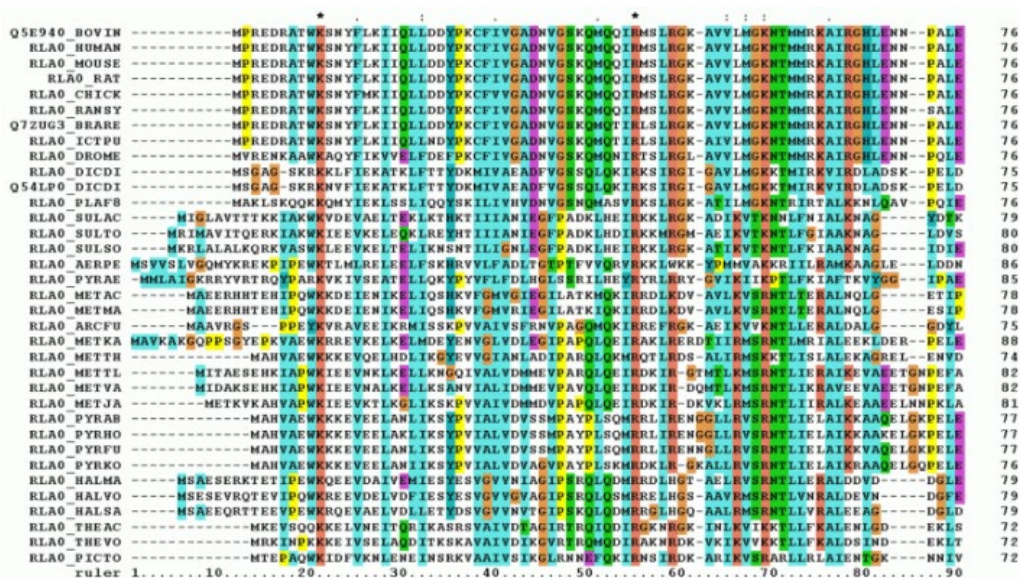


Figure 2: for an example of a multiple alignment output

**PROBLEM**

One of the first programs used for MSA was Clustal, developed by Des Higgins in 1988. ClustalW2 is the classic version that is embedded in many software packages. ClustalX is a modified version with a graphical user interface. Clustal Omega is the latest version with a significant increase in scalability and quality of alignments due to a new HMM alignment engine.

See the link below for a convenient online interface that runs Clustal Omega on the [EBI](#) website:

[Clustal Omega](#)

Given: A file containing a set of nucleotide strings in FASTA format.

Return: ID of the string most different from the others.

Note: instead of code, please submit a screenshot of the phylogram on the Clustal Omega Guide Tree webpage.

Sample Dataset

```
>Rosalind_18
GACATGTTTGTTCCTTAACTCGTGGCGGCCTAGCCGTAAGTTAAG
>Rosalind_23
ACTCATGTTTGTTCCTTAACTCTTGGCGGCTTAGCCGTAAGTTAAG
>Rosalind_51
TCCTATGTTTGTTCCTCAAACCTTGGCGGCCTAGCCGTAAGGTAAG
>Rosalind_7
CACGTCTGTTGCGCTAAACCTTTGATTGCCGGCCTACGCTAGTTAGTTA
>Rosalind_28
GGGGTCATGGCTGTTTGCCTTAAACCCTTGGCGGCCTAGCCGTAATGTTT
```

Sample Output

Rosalind_7

Hint: you may run Clustal Omega via the command line like done in Homework 5. You may also run it via a Biopython wrapper. Warning: Biopython has decided to drop MSA command line wrappers in a future Biopython release. The [documentation](#) will be updated to instead build the command line directly and invoke it with the subprocess module.



2. Overlap Graphs

Networks arise everywhere in the practical world, especially in biology. Networks are prevalent in popular applications such as modeling the spread of disease, but the extent of network applications spreads far beyond popular science. Our first question asks how to computationally model a network without needing to render a picture of the network.

First, some terminology: graph is the technical term for a network; a graph is made up of hubs called nodes (or vertices), pairs of which are connected via segments/curves called edges. If an edge connects nodes v and w , then it is denoted by v, w (or equivalently w, v).

- an edge v, w is incident to nodes v and w ; we say that v and w are adjacent to each other;
- the degree of v is the number of edges incident to it;
- a walk is an ordered collection of edges for which the ending node of one edge is the starting node of the next (e.g., $\{v1, v2\}$, $\{v2, v3\}$, $\{v3, v4\}$, etc.);
- a path is a walk in which every node appears in at most two edges;
- path length is the number of edges in the path;
- a cycle is a path whose final node is equal to its first node (so that every node is incident to exactly two edges in the cycle); and
- the distance between two vertices is the length of the shortest path connecting them.

Graph theory is the abstract mathematical study of graphs and their properties.

PROBLEM

A graph whose nodes have all been labeled can be represented by an adjacency list, in which each row of the list contains the two node labels corresponding to a unique edge.

A directed graph (or digraph) is a graph containing directed edges, each of which has an orientation. That is, a directed edge is represented by an arrow instead of a line segment; the starting and ending nodes of an edge form its tail and head, respectively. The directed edge with tail v and head w is represented by (v, w) (but not by (w, v)). A directed loop is a directed edge of the form (v, v) .

For a collection of strings and a positive integer k , the overlap graph for the strings is a directed graph O_k in which each string is represented by a node, and string s is connected to string t with a directed edge when there is a length k suffix of s that matches a length k prefix of t , as long as $s \neq t$; we demand $s \neq t$ to prevent directed loops in the overlap graph (although directed cycles may be present).

Given: A file containing a collection of DNA strings in FASTA format having total length at most 10 kb.

Return: The adjacency list corresponding to O_3 . You may return edges in any order.



Sample Dataset

```
>Rosalind_0498
AAATAAA
>Rosalind_2391
AAATTTT
>Rosalind_2323
TTTTCCC
>Rosalind_0442
AAATCCC
>Rosalind_5013
GGGTGGG
```

Sample Output

```
Rosalind_0498 Rosalind_2391
Rosalind_0498 Rosalind_0442
Rosalind_2391 Rosalind_2323
```

Hint: If you are looking for a way to actually visualize graphs as you are working through the homework, then you may like to consider [Graphviz](#), a cross-platform application for rendering graphs.

3. Completing a Tree/Counting Phylogenetic Ancestors

"As buds give rise by growth to fresh buds, and these, if vigorous, branch out and overtop on all sides many a feebler branch, so by generation I believe it has been with the great Tree of Life, which fills with its dead and broken branches the crust of the earth, and covers the surface with its ever-branching and beautiful ramifications."
-Charles Darwin, *The Origin of Species*

A century and a half has passed since the publication of Darwin's magnum opus, and yet the construction of a single Tree of Life uniting life on Earth still has not been completed, with perhaps as many as 90% of all living species not yet catalogued (although a beautiful interactive animation has been produced by [OneZoom](#)).

To learn about state-of-the-art attempts to build the Tree of Life, you may like to learn about the [Tree of Life Web Project](#), a collaborative effort by biologists from around the world. This peer-reviewed project started in 1995, and today it has grown to contain more than 10,000 pages with characteristics of different groups of organisms and their evolutionary history.

Instead of trying to construct the entire Tree of Life all at once, we often wish to form a simpler tree in which a collection of species have been clumped together for the sake of simplicity; such a group is called a taxon (plural taxa). For a given collection of taxa, a phylogeny is a treelike diagram that best represents the evolutionary connections between the taxa: the construction of a particular phylogeny depends on our specific assumptions regarding how these evolutionary relationships should be interpreted. See Figure 1.

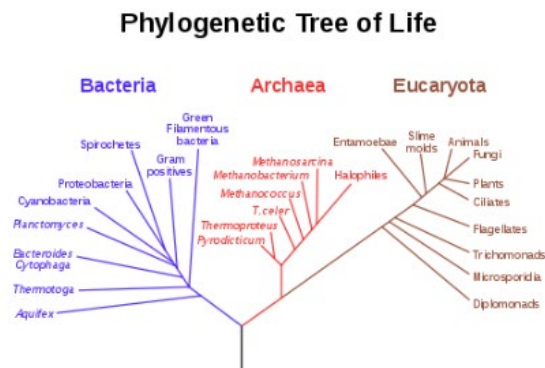


Figure 1: A phylogeny illustrating proposed evolutionary relationships among the three domains of life: Bacteria, Archaea, and Eukaryota.

An undirected graph is connected if there is a path connecting any two nodes. A tree is a connected (undirected) graph containing no cycles; this definition forces the tree to have a branching structure organized around a central core of nodes, just like its living counterpart. See Figure 2.

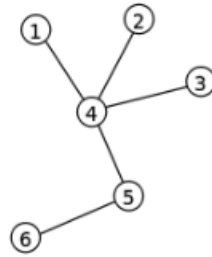


Figure 2: A labeled tree with 6 vertices and 5 edges.

In the creation of a phylogeny, taxa are encoded by the tree's leaves, or nodes having degree 1. A node of a tree having degree larger than 1 is called an internal node.

The definition of tree as a connected graph with no cycles produces a huge class of different graphs, from simple paths and star-like graphs to more familiar arboreal structures (see Figure 3). Which of these graphs are appropriate for phylogenetic study?

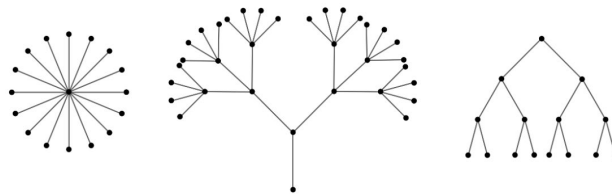


Figure 3: Trees come in lots of different shapes.

Modern evolutionary theory (beginning with Darwin) indicates that a common way for a new species to be created is when it splits off from an existing species after a population is isolated for an extended period. This model of species evolution implies a very specific type of phylogeny, in which internal nodes represent branching points of evolution where an ancestor species either evolved into a new species or split into two new species: therefore, one edge of this internal node connects the node to its most recent ancestor, whereas one or two new edges connect it to its immediate descendants. This framework offers a much clearer notion of how to characterize phylogenies.

PROBLEM

A binary tree is a tree in which each node has degree equal to at most 3. The binary tree will be our main tool in the construction of phylogenies.

A rooted tree is a tree in which one node (the root) is set aside to serve as the pinnacle of the tree. A standard graph theory exercise is to verify that for any two nodes of a tree, exactly one path connects the nodes. In a rooted tree, every node v will therefore have a single parent, or the unique node w such that the path from v to the root contains $\{v, w\}$. Any other node x adjacent to v is called a child of v because v must be the parent of x ; note that a node may have multiple children. In other words, a rooted tree possesses an ordered hierarchy from the root down to its leaves, and as a result, we may often view a rooted tree with undirected edges as a directed graph in which each edge is oriented from parent to child.



Even though a binary tree can include nodes having degree 2, an unrooted binary tree is defined more specifically: all internal nodes have degree 3. In turn, a rooted binary tree is such that only the root has degree 2 (all other internal nodes have degree 3). See Figure 4 for the difference between a rooted and unrooted binary tree.

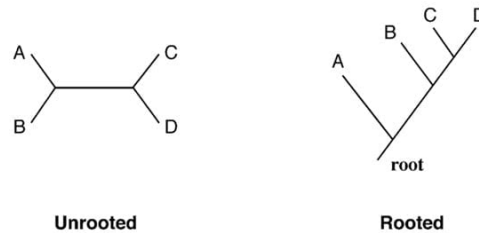


Figure 4: Unrooted vs. rooted binary tree.

Given: A file containing a positive integer n ($n \leq 1000$) and an adjacency list corresponding to a graph of n nodes that contains no cycles.

Return: The minimum number of edges that can be added to the graph to produce a tree, and the number of internal nodes of any unrooted binary tree having n leaves.

Sample Dataset

```
10
1 2
2 8
4 10
5 9
6 10
7 9
```

Sample Output

```
3 8
```

Hint: The coding part of this problem is trivial. For the first question, consider Figure 2 and the number of edges required for a tree with six nodes. Then, generalize this concept for a tree with n nodes. The problem is simply asking how many more edges must be added to arrive at the correct total number of edges when you are already given a certain number of edges in the input dataset. For the second question, consider Figure 4 and the number of internal nodes required for an unrooted tree with four leaves. Then, generalize this concept for an unrooted tree with n leaves.



4. Creating a Distance Matrix

Several different approaches are used to build phylogenies, each one featuring its own computational strengths and weaknesses. One of these measures is distance-based phylogeny, which constructs a tree from evolutionary distances calculated between pairs of taxa.

A wide assortment of different measures exist for quantifying this evolutionary distance. Once you have selected a distance function and used it to calculate the distance between every pair of taxa, you place these pairwise distance functions into a table.

In this problem, you will consider an evolutionary function based on the Hamming distance. Recall from the “HAMM: Counting Point Mutations” problem that this function compares two homologous strands of DNA by counting the minimum possible number of point mutations that could have occurred on the evolutionary path between the two strands.

PROBLEM

For two strings s_1 and s_2 of equal length, the p-distance between them, denoted $d_p(s_1, s_2)$, is the proportion of corresponding symbols that differ between s_1 and s_2 .

For a general distance function d on n taxa $s_1, s_2, \dots, s_n$ (taxa are often represented by genetic strings), we may encode the distances between pairs of taxa via a distance matrix D in which $D_{i,j} = d(s_i, s_j)$.

Given: A collection of n ($n \leq 10$) DNA strings $s_1, s_2, \dots, s_n$ of equal length (at most 1 kb). Strings are given in FASTA format.

Return: A file containing a matrix D corresponding to the p-distance d_p on the given strings. Note that your answer is allowed an absolute error of 0.001.

Sample Dataset

```
>Rosalind_9499
TTTCCATTTA
>Rosalind_0942
GATTCATTTC
>Rosalind_6568
TTTCCATTTT
>Rosalind_1833
GTTCCATTTA
```

Sample Output

```
0.00000 0.40000 0.10000 0.10000
0.40000 0.00000 0.40000 0.30000
0.10000 0.40000 0.00000 0.20000
0.10000 0.30000 0.20000 0.00000
```

Hint: Import the FASTA file, then loop through and calculate the distances for each pair of sequences to create the distance matrix.